

The Bifurcated Equiprobability Theory: A Comprehensive Re-examination of All Binary Outcomes

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Abstract

Every measurable event determines a binary alternative between occurrence and nonoccurrence. Starting from normalization and complement consistency, without assuming additivity, we derive a universal equal-half assignment for proper events through label-interchange invariance. The resulting assignment is monotone but nonadditive on event spaces containing three disjoint nonempty events. We examine the consequences for repeated observations within the same framework. Counts no longer obey binomial aggregation, deviation events retain assignment one half as the number of observations increases, and standardized sums fail to approach the classical Gaussian distribution function. Almost-sure frequency convergence likewise fails on the infinite sequence space. Taken together, these results present a structural challenge to the classical additive framework: binary normalization remains available, while the aggregation and limiting principles built upon global additivity cease to hold.

1 Introduction

The persistent quest to identify the basic structure underlying random phenomena has led to the formulation of a wide variety of probability distributions. In the present work we propose a unifying alternative, which we call *Bifurcated Equiprobability Theory* (BET). The central observation of BET is that every probabilistic assertion can be reduced to a binary alternative: either a given event occurs, or its complement occurs.

Thus, rather than beginning with a prescribed probability distribution on a sample space, BET begins with the event-complement pair associated with a measurable event. If E is an event, then the relevant bifurcation is

$$\{E, E^c\}.$$

This pair consists of two mutually exclusive and exhaustive alternatives. BET asserts that this

bifurcated structure already determines the associated probability law. The reduction theorem below assigns probability $\frac{1}{2}$ to each alternative.

In this sense, the Coin-Flip (CF) distribution is not introduced as a special probability distribution attached to a particular random mechanism. Rather, it appears as the universal reduced law associated with any event once that event is expressed in its binary form. The role of BET is therefore to show that arbitrary probabilistic models collapse, event by event, to the same bifurcated equiprobable structure.

A probabilistic question often has a binary form even when the underlying model is highly structured. A measurement either exceeds a threshold or does not; a selected outcome either belongs to a prescribed set or lies outside it; an observed increment is either positive or nonpositive. Such questions retain only the distinction between an event and its complement.

Their formulation does not require the original sample space to be finite, discrete, or equipped with a topology.

The usual measure-theoretic construction expresses this reduction through an indicator random variable. Starting with a probability space, one transfers its measure to a two-element target by taking inverse images. This construction is standard; see, for example, [1]. The classical axiomatization places additivity at its foundation [5,6]. The present work weakens the starting structure: we retain normalization and complement consistency but do not assume additivity across arbitrary disjoint events. Nonadditive set functions have also been studied in capacity theory and in nonadditive integration [7,8]. We then address the law selected by the binary reduction itself. We call the proposed framework *Bifurcated Equiprobability Theory* (BET), and its distinguished reduced distribution the *Coin-Flip law*.

Our main result identifies every proper-event binary law with the uniform measure on the two-element target. Its argument uses the interchange of the two outcome labels. The claimed universality has a stronger implication than the construction of one nonadditive example: it excludes an additive assignment on any event algebra containing three pairwise disjoint nonempty events. We develop this implication as a challenge to the classical probability framework itself. Earlier work on subjective probability and ambiguity has considered alternative assessments of uncertainty [9–12]; those references motivate comparison, rather than supply a proof of BET’s universal conclusion.

The same assignment is used throughout the analysis of repetitions. No additive product measure is introduced to restore binomial sampling. Instead, events on finite and infinite sequence spaces are evaluated by their own event-complement bifurcations. This makes it possible to examine directly whether increasing the number of observations produces the concentration and Gaussian limits predicted by classical theory [1, 13, 14].

Section 2 fixes the initial structure. Section 2.2 establishes the binary reduction and derives non-

additivity. Section 3 records local binary quantities. Section 4 studies counts, frequencies, and standardized sums. Section 5 applies BET to the textbook example of birth rate prediction. Section 6 shows how relevant stocks follow the presented probability distribution. Section 7 gives further applications, and Section 8 examines the consequences for classical statistical assessment.

2 Theory

2.1 Binary observables and induced laws

Let $(\Omega, \mathcal{F})$ be a measurable space with $\Omega \neq \emptyset$, and let $\mathbf{2} = \{0, 1\}$ carry its power-set σ -algebra. Complements are always taken in Ω .

Definition 1 (Complement-consistent assignment). A function $Q : \mathcal{F} \rightarrow [0, 1]$ is a complement-consistent assignment if

$$\begin{aligned} Q(\emptyset) &= 0, & Q(\Omega) &= 1, \\ Q(E^c) &= 1 - Q(E), & E &\in \mathcal{F}. \end{aligned}$$

We call $(\Omega, \mathcal{F}, Q)$ a bifurcation space. Neither finite nor countable additivity is assumed.

Complement consistency normalizes each event-complement pair. It does not specify the assignment to the union of two arbitrary disjoint events. Every ordinary probability measure satisfies these initial conditions, but the class also allows nonadditive assignments.

Definition 2 (Proper event). An event $E \in \mathcal{F}$ is *proper* if

$$\emptyset \neq E \neq \Omega.$$

This is a set-theoretic condition; it does not initially impose a restriction on $Q(E)$.

Definition 3 (Event-complement reduction). For an event E , its binary observable and induced law are

$$\begin{aligned} \beta_E(\omega) &= \mathbf{1}_E(\omega), \\ \nu_E(A) &= Q(\beta_E^{-1}(A)), \quad A \subseteq \mathbf{2}. \end{aligned}$$

Equivalently, $\nu_E = (\beta_E)_*Q$ is the set function obtained by taking inverse images. For proper E , the family $\mathcal{B}_E = \{E, E^c\}$ is its event-complement bifurcation.

Measurability follows because the inverse images of the two singletons are E and E^c . In particular,

$$\begin{aligned}\nu_E(\{1\}) &= Q(E), \\ \nu_E(\{0\}) &= Q(E^c).\end{aligned}\tag{1}$$

For proper E , the map β_E is surjective. Although Q need not be additive on $\mathcal{F}$, ν_E is an ordinary probability measure on $\mathbf{2}$. Indeed, its singleton masses are nonnegative and sum to one by complement consistency; its values at $\emptyset$ and $\mathbf{2}$ are zero and one. These four values verify countable additivity on the finite target. This conclusion holds for degenerate events as well.

Definition 4 (Coin-Flip law). The Coin-Flip law is the measure ν_{CF} on $\mathbf{2}$ satisfying

$$\nu_{\text{CF}}(\{0\}) = \nu_{\text{CF}}(\{1\}) = \frac{1}{2}.$$

Thus $\nu_{\text{CF}} = \text{Bern}(1/2)$.

Let $\tau : \mathbf{2} \rightarrow \mathbf{2}$ be the label interchange $\tau(b) = 1 - b$. It satisfies $\tau^{-1} = \tau$ and

$$\begin{aligned}\beta_{E^c} &= \tau \circ \beta_E, \\ \nu_{E^c} &= \tau_* \nu_E.\end{aligned}$$

These identities describe how the reduced law transforms when occurrence and nonoccurrence exchange labels.

Lemma 1 (Invariant binary measure). *A probability measure ν on $\mathbf{2}$ satisfies $\tau_* \nu = \nu$ if and only if $\nu = \nu_{\text{CF}}$.*

Proof. Invariance evaluated on $\{1\}$ gives

$$\nu(\{1\}) = \nu(\tau^{-1}(\{1\})) = \nu(\{0\}).$$

The two masses sum to one, so both equal $1/2$. Conversely, equal masses are preserved by their interchange. $\square$

2.2 Universal binary reduction

Theorem 1 (Universal binary reduction). *For every bifurcation space $(\Omega, \mathcal{F}, Q)$ and every proper event $E \in \mathcal{F}$, one has*

$$(\beta_E)_*Q = \nu_{\text{CF}}.$$

In particular, $Q(E) = Q(E^c) = 1/2$.

Proof. The proper event E gives two nonempty, disjoint fibers whose union is Ω . By normalization and complement consistency, its induced law is a probability measure on the binary target. The involution τ interchanges the two labels while preserving the event-complement structure. Since this interchange changes only the labeling of the two alternatives, the induced probability measure is invariant under it:

$$\tau_* \nu_E = \nu_E.\tag{2}$$

Lemma 1 now gives $\nu_E = \nu_{\text{CF}}$. Evaluating the two singleton masses using (1) proves the final assertion. $\square$

Remark 1 (Degenerate events). The theorem excludes $\emptyset$ and Ω . Their binary observables are constant, with respective laws δ_0 and δ_1 . Their assignments remain zero and one.

Corollary 1 (Measurable predicates). *Let $X : \Omega \rightarrow S$ be measurable, where $(S, \mathcal{S})$ is a measurable space. If $A \in \mathcal{S}$ has a proper inverse image $X^{-1}(A)$, then*

$$Q(X \in A) = Q(X \notin A) = \frac{1}{2}.$$

Proof. Apply Theorem 1 to $E = X^{-1}(A)$, using $E^c = X^{-1}(S \setminus A)$. $\square$

The condition concerns the inverse image, not merely the target set A . A proper subset of S may contain all values of X , or none of them.

Corollary 2 (Global assignment and nonadditivity). *Under Theorem 1, the assignment is*

$$Q(E) = \begin{cases} 0, & E = \emptyset, \\ 1, & E = \Omega, \\ 1/2, & \emptyset \neq E \neq \Omega. \end{cases}$$

This assignment is monotone and complement-consistent. If $\mathcal{F}$ contains three pairwise disjoint nonempty events, it is not finitely additive.

Proof. The formula combines normalization with Theorem 1. Complement consistency follows by separating degenerate and proper events. If $A \subseteq B$, either $A = \emptyset$, or $B = \Omega$, or both are proper. The displayed values give $Q(A) \leq Q(B)$ in every case.

Let A, B, C be pairwise disjoint nonempty events. Both A and B are proper, and $A \cup B$ is proper because its complement contains C . Hence

$$\begin{aligned} Q(A \cup B) &= \frac{1}{2}, \\ Q(A) + Q(B) &= 1. \end{aligned}$$

This violates finite additivity. $\square$

For example, on the full event algebra of $\Omega = \{1, 2, 3\}$, each singleton and each two-element subset receives assignment $1/2$. This is compatible with complement consistency: each singleton is paired with its two-element complement. The failure occurs when two disjoint singletons are combined.

Remark 2 (Local and global additivity). For a proper event E , the restriction of Q to

$$\mathcal{F}_E = \{\emptyset, E, E^c, \Omega\}$$

is an ordinary probability measure. Global non-additivity concerns combinations of events beyond a single such algebra. Thus the conclusion concerns the extension of additive reasoning across distinct binary questions, rather than normalization within one question.

3 Consequences for one binary observable

Fix a proper event E . On the ordinary probability space $(\mathbf{2}, 2^{\mathbf{2}}, \nu_E)$, let $B(b) = b$ be the canonical coordinate. Its law is the binary law of β_E , which Theorem 1 identifies with ν_{CF} . Expectations and variances in this section are taken on this binary space. Equivalently, they concern

the additive restriction to $\mathcal{F}_E$; no integral with respect to Q on the full algebra is introduced.

Proposition 1 (Expectation and two-level responses). *For every function $f : \mathbf{2} \rightarrow \mathbb{R}$,*

$$\mathbb{E}[f(B)] = \frac{f(0) + f(1)}{2}.$$

Consequently, for $Y = a + (b - a)B$, with $a, b \in \mathbb{R}$,

$$\begin{aligned} \mathbb{E}[Y] &= \frac{a + b}{2}, \\ \text{Var}(Y) &= \frac{(b - a)^2}{4}. \end{aligned}$$

Proof. The expectation formula is the sum over the two atoms of ν_{CF} . In particular, $\mathbb{E}[B] = \mathbb{E}[B^2] = 1/2$, so $\text{Var}(B) = 1/4$. Affine transformation gives the remaining identities. $\square$

Thus the variance depends on the distance between the two response levels, even though the binary law itself is fixed. Equal probabilities do not imply equal response magnitudes.

It is sometimes convenient to use the centered encoding $Z = 2B - 1$. Its values are -1 and 1 , and

$$\begin{aligned} \mathbb{E}[Z] &= 0, & \text{Var}(Z) &= 1, \\ \mathbb{E}[e^{tZ}] &= \cosh t, & t &\in \mathbb{R}. \end{aligned}$$

All odd moments of Z vanish and all even moments equal one.

Proposition 2 (Binary information quantities). *Using logarithms to base two, the entropy of B is one bit. If $q = \text{Bern}(p)$ is any comparison law, then*

$$\begin{aligned} D(q \| \nu_{\text{CF}}) &= 1 - h_2(p), \\ d_{\text{TV}}(q, \nu_{\text{CF}}) &= |p - 1/2|, \end{aligned}$$

where

$$h_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

and $0 \log_2 0$ is interpreted as zero.

Proof. For ν_{CF} , the two entropy contributions are each $1/2$. By the definition of relative entropy,

$$\begin{aligned} D(q\|\nu_{\text{CF}}) &= p \log_2(2p) \\ &\quad + (1-p) \log_2(2(1-p)) \\ &= 1 - h_2(p). \end{aligned}$$

On a finite space, total variation is half the sum of the absolute differences of the atomic masses. Here this gives

$$d_{\text{TV}}(q, \nu_{\text{CF}}) = \frac{1}{2}(|p - 1/2| + |1/2 - p|),$$

which proves the claim, including the endpoints. $\square$

These are standard information-theoretic definitions [2]. The comparison law q can represent a fitted binary model or an empirical distribution. The identities quantify its departure from the Coin-Flip reference without identifying finite-sample frequencies with population probabilities.

4 Repeated observations and classical limits

For $n \geq 1$, let $\Omega_n = \mathbf{2}^n$ with its full event algebra. Equip it with a normalized, complement-consistent assignment Q_n . Theorem 1 gives

$$Q_n(A) = \frac{1}{2}, \quad \emptyset \neq A \neq \Omega_n.$$

Let $B_i(\omega) = \omega_i$ be the coordinate observables and define

$$K_n = \sum_{i=1}^n B_i, \quad \hat{p}_n = \frac{K_n}{n}.$$

The coordinate notation describes repetitions without imposing a classical product law. In particular, the identity $Q_n(A \cap B) = Q_n(A)Q_n(B)$ is not an initial assumption.

Proposition 3 (Failure of binomial aggregation). *For every $n \geq 1$ and $k \in \{0, \dots, n\}$,*

$$Q_n(K_n = k) = \frac{1}{2}.$$

For $n \geq 2$, these event assignments do not form a probability mass function on the count values.

Proof. Every count k is realized by at least one binary sequence. A different count is also realized, so $\{K_n = k\}$ is proper. Theorem 1 applies. For $n \geq 2$, summing the assignments gives $(n+1)/2 > 1$, whereas the assignment to their union is one. This discrepancy is precisely the failure of finite additivity. $\square$

For instance, with two observations the three count events have assignments

$$Q_2(K_2 = 0) = Q_2(K_2 = 1) = Q_2(K_2 = 2) = \frac{1}{2}.$$

The classical independent fair-trial weights would instead be $1/4$, $1/2$, and $1/4$ [13]. They arise by summing product weights over elementary sequences, an operation incompatible with the global assignment above.

Proposition 4 (Failure of frequency concentration). *Fix $0 < \varepsilon < 1/2$. For all sufficiently large n ,*

$$Q_n(|\hat{p}_n - 1/2| \geq \varepsilon) = \frac{1}{2}.$$

Consequently, the classical concentration bound

$$Q_n(|\hat{p}_n - 1/2| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}$$

fails for all sufficiently large n .

Proof. The all-zero sequence belongs to the deviation event, since its deviation is $1/2$. Choose an integer k_n nearest to $n/2$. A sequence with k_n ones satisfies

$$|k_n/n - 1/2| \leq \frac{1}{2n}.$$

For sufficiently large n , this is less than ε , so the deviation event is proper. Its assignment is therefore $1/2$. The proposed upper bound tends to zero and is strictly below $1/2$ once $n > \log(4)/(2\varepsilon^2)$. $\square$

If convergence in assignment is defined by the vanishing of the assignment to each fixed positive-deviation event, the empirical frequencies therefore do not converge to $1/2$ in that

sense. The obstruction persists with arbitrarily many observations; it is not a finite-sample correction.

Proposition 5 (Failure of the classical Gaussian limit). *Define the classically standardized count*

$$Z_n = \frac{2K_n - n}{\sqrt{n}}.$$

For every fixed $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} Q_n(Z_n \leq x) = \frac{1}{2}.$$

In particular, these quantities do not converge to the standard normal distribution function $\Phi(x)$ at every continuity point.

Proof. The minimum and maximum of Z_n are $-\sqrt{n}$ and $\sqrt{n}$. For any fixed x , sufficiently large n satisfy

$$-\sqrt{n} \leq x < \sqrt{n}.$$

The all-zero sequence then belongs to $\{Z_n \leq x\}$ and the all-one sequence does not. This event is proper, so its assignment is $1/2$. To distinguish the limit from a standard Gaussian, take $x = 1$: positivity of the Gaussian density on $(0, 1)$ gives $\Phi(1) > \Phi(0) = 1/2$. $\square$

The expression $Q_n(Z_n \leq x)$ is an event-assignment profile. It need not be a classical cumulative distribution function because no global additivity is available. The proposition concerns the explicit Gaussian-limit formula used in the classical central limit theorem [1, 6]; it does not invoke a weak-convergence theorem for nonadditive set functions. The scaling is the classical one, not a variance computed by integrating K_n against Q_n .

Proposition 6 (Failure of almost-sure frequency convergence). *Let $\Omega_\infty = \mathbf{2}^{\mathbb{N}}$ with its product σ -algebra, equipped with the equal-half assignment Q_∞ . The event*

$$L = \{\omega : \lim_{n \rightarrow \infty} \hat{p}_n(\omega) = 1/2\}$$

satisfies $Q_\infty(L) = Q_\infty(L^c) = 1/2$.

Proof. The event is measurable because it can be written as

$$L = \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{|\hat{p}_n - 1/2| < 1/m\}.$$

The alternating sequence belongs to L , while the all-zero sequence does not. Thus L is proper, and both assignments equal $1/2$. $\square$

In particular, the convergence event does not have assignment one and its complement does not have assignment zero. The assignment analogue of the classical almost-sure conclusion fails. Finite and infinite assignments agree on cylinder events: the inverse image of a proper finite-sequence event under coordinate projection is again proper.

5 Validation

In order to examine the empirical interpretation of the CF distribution, we return to one of the original examples: births recorded in two categories, traditionally denoted “male” and “female”. The binary question is defined on records classified in those two categories; no other scenario can happen as an outcome of a birth, and the classification assigned the first moments of life is constant throughout life until senility.

Let E denote the male-coded category in this sample space, with both categories nonempty. The CF assignment is

$$Q(E) = Q(E^c) = \frac{1}{2}.$$

For a recorded population, however, its male proportion is the arithmetic quantity

$$\hat{p} = \frac{N_M}{N_M + N_F}.$$

If r denotes male births per 100 female births, the corresponding proportion is

$$\hat{p} = \frac{r}{100 + r}.$$

Consequently, ratios of 105, 106, and 107 correspond respectively to approximately 51.22%,

51.46%, and 51.69%. These values must not be rounded to exact equality with 50%.

Table 1 compares the CF assignment with published demographic ratios. The US historical series was documented by the National Center for Health Statistics [3]; the World Bank reports 2023 ratios for Afghanistan and Albania [4]; and UNFPA reports the Chinese 2020 census ratio [15]. For orientation, the often-used baseline of 105 male births per 100 female births is shown separately as a numerical reference, rather than presented as a measured worldwide total.

The interpretation of these comparisons must be consistent with Section 4. BET does not imply convergence of the recorded fraction to $1/2$: that frequency-limit conclusion has itself been rejected. Likewise, although each individual binary marginal has mean $1/2$, summing these marginal means does not define a global expectation of the count under the nonadditive assignment. Thus the formula $\mathbb{E}[N_M] = N/2$ cannot be imported as a statement about a global BET expectation without additional structure.

The demographic example therefore illustrates the distinction between a binary assignment and an observed population fraction. A discrepancy between them is not removed by rounding, and proximity to one half does not by itself validate BET. An empirical test would require an operational link between Q and observations beyond the structural results established here.

6 Empirical Illustration of the Coin-Flip Distribution in Market Trajectories

The second original example concerns financial trajectories viewed over three different time horizons. For an observed index level S_i , define the increment $R_i = S_i - S_{i-1}$. The binary question is whether the increment is positive or nonpositive. If its event is proper, the reduction gives

$$Q(R_i > 0) = Q(R_i \leq 0) = \frac{1}{2}.$$

Zero increments belong to the second branch. A positive-versus-negative partition would instead

require their exclusion from the underlying sample space.

Figures 1–3 display three NASDAQ trajectories over long, intermediate, and short horizons, respectively. The index also provides historical series through its official data portal [16].

An endpoint-based argument sometimes invokes Rolle’s theorem to relate endpoint agreement to balance between upward and downward movements. Its precise implication is narrower: for a function continuous on $[a, b]$, differentiable on (a, b) , and satisfying $S(a) = S(b)$, there is a point $c \in (a, b)$ with $S'(c) = 0$. This does not establish equal counts or equal probabilities of positive and negative increments.

The relevant identity for cumulative magnitudes is instead a telescoping sum. For a discrete path $S_0, \dots, S_N$, set

$$U = \sum_{i=1}^N \max\{R_i, 0\},$$

$$D = \sum_{i=1}^N \max\{-R_i, 0\}.$$

Then

$$U - D = \sum_{i=1}^N R_i = S_N - S_0.$$

Equal endpoints therefore imply equal cumulative upward and downward magnitudes. Approximately equal endpoints imply that their difference is correspondingly small in absolute units. Neither conclusion determines how many increments have each sign: increments $2, -1, -1$ already give equal endpoints with unequal positive and negative counts.

For an absolutely continuous interpolation, the analogous identity is

$$S(b) - S(a) = \int_a^b S'(t) dt$$

$$= \int_a^b (S'(t))_+ dt - \int_a^b (S'(t))_- dt,$$

where $v_+ = \max\{v, 0\}$ and $v_- = \max\{-v, 0\}$. This balances total variation components when

Table 1: Selected reported birth sex ratios and the theoretical CF assignment. The first row is an illustrative baseline, not a country-year observation; percentages are calculated as $100r/(100 + r)$.

Population / period	Male per 100 female	Male share	CF assignment
Illustrative baseline	105	51.22%	50%
United States (1946) [3]	105.9	51.43%	50%
China (2020 census) [15]	111.3	52.67%	50%
Afghanistan (2023) [4]	105	51.22%	50%
Albania (2023) [4]	107	51.69%	50%



Figure 1: Long-period NASDAQ trajectory over the indicated observation window.

endpoints agree, not the duration or probability of either direction.

The three graphical horizons consequently provide concrete settings for the same binary question. Under BET their event assignments come from Theorem 1, independently of a visual impression of balance. Neither oscillation nor endpoint coincidence constitutes an additional proof of equiprobability, and no frequency convergence is inferred from the short-term example.

7 Analytical applications

The following examples use the same global assignment. They are analytical consequences rather than empirical observations.

7.1 Finite selection

Let $\Omega = \{1, \dots, m\}$ with its full event algebra. Every nonempty proper subset receives assignment $1/2$, independently of its cardinality. For $m \geq 3$, the result differs from uniform additive counting, which assigns $|A|/m$ to a subset A . Under the universal reduction theorem, the discrepancy concerns the aggregation rule itself.

7.2 Threshold events

Let $X : \Omega \rightarrow \mathbb{R}$ be measurable. Whenever $\{X > t\}$ is proper,

$$Q(X > t) = Q(X \leq t) = \frac{1}{2}.$$

Values equal to the threshold belong to the second branch. Omitting them would destroy exhaustiveness whenever that level set is nonempty.

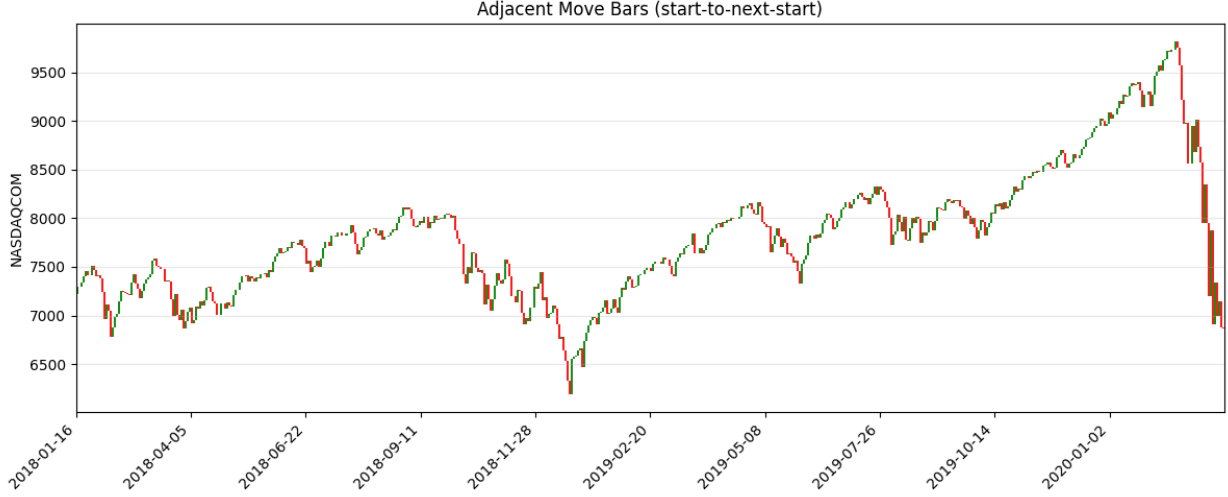


Figure 2: Medium-period NASDAQ evolution over the indicated observation window.

For the identity observable on $\mathbb{R}$ with its Borel algebra, both half-lines are proper for every finite threshold. Thus

$$Q((-\infty, t]) = \frac{1}{2}, \quad t \in \mathbb{R}.$$

This profile has neither of the classical limits zero at $-\infty$ and one at $+\infty$. Its behavior is consistent with the failure of continuity under monotone limits established below.

7.3 Monotone sequences of events

On $\mathbb{N}$ with its full event algebra, set $A_n = \{1, \dots, n\}$. Then

$$A_n \uparrow \mathbb{N}, \quad Q(A_n) = \frac{1}{2},$$

$$Q\left(\bigcup_n A_n\right) = 1.$$

Similarly, $A_n^c \downarrow \emptyset$ while $Q(A_n^c) = 1/2$ for every n . Thus the equal-half assignment fails both continuity from below and continuity from above. Monotonicity of the set function does not imply either continuity property.

7.4 Directional observations

For a measurable increment R with proper positive-sign event,

$$Q(R > 0) = Q(R \leq 0) = \frac{1}{2}.$$

This assignment records direction only. It does not define an expectation of R under a globally additive integral, and it does not assert equal magnitudes of gains and losses. A two-level response attached to the sign question can still be evaluated on its binary algebra by Proposition 1. General magnitude information requires additional structure.

8 Consequences for statistical assessment

Classical significance calculations depend on assigning probabilities to collections of possible samples. The event-level assignment derived here changes those calculations even when the sample space and rejection criterion are left unchanged.

Proposition 7 (Assignments to rejection regions). *Let $\mathcal{R} \subset \Omega_n$ be a deterministic rejection region for a test based on n binary observations. If both rejection and nonrejection are possible, then*

$$Q_n(\mathcal{R}) = Q_n(\mathcal{R}^c) = \frac{1}{2}.$$

Hence no such region has BET rejection assignment at most α for $0 < \alpha < 1/2$.

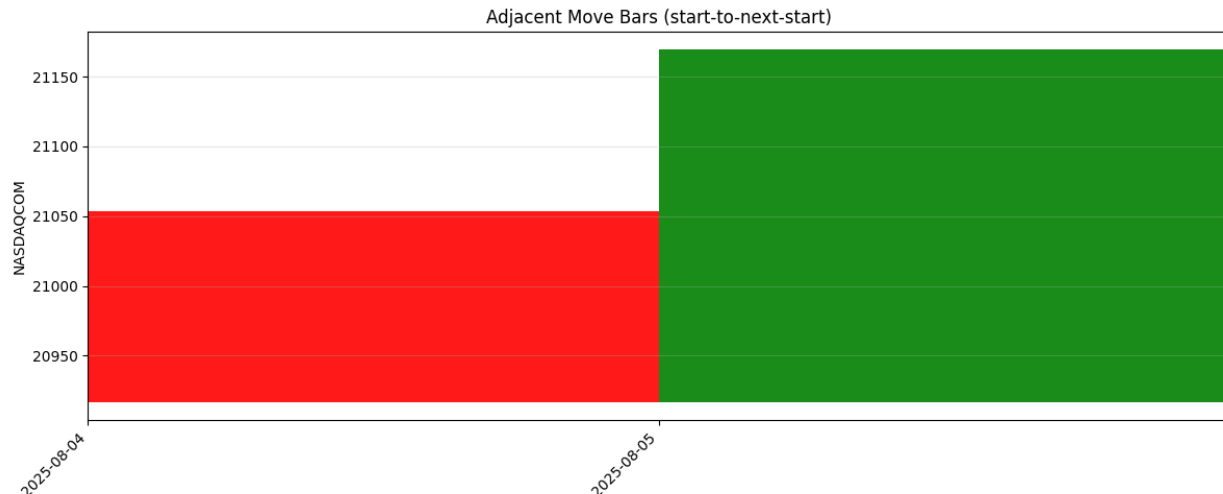


Figure 3: Short-term NASDAQ fluctuation over the indicated observation window.

Proof. The hypothesis states exactly that $\mathcal{R}$ is proper. Apply Theorem 1. $\square$

For a concrete comparison, let $n = 10$ and consider

$$\mathcal{R} = \{K_{10} \leq 1\} \cup \{K_{10} \geq 9\}.$$

Under the classical independent fair-trial model, this event has probability

$$2^{-10} \sum_{j \in \{0,1,9,10\}} \binom{10}{j} = \frac{22}{1024}.$$

Under BET its assignment is $1/2$, since some binary sequences belong to $\mathcal{R}$ and others do not. The classical calculation is an exact calculation in its own additive model; the universal reduction challenges that model's admissibility as a global event assignment.

These are mathematical comparisons, not empirical confirmations. The demographic ratios and graphical examples above do not supply a frequency-calibration rule for Q . Any empirical interpretation would additionally need an operational connection between the set function and observations. In particular, frequency calibration cannot simply be assumed after the classical frequency-limit conclusions have been rejected.

9 Discussion

The distinction between a nonadditive example and a universal reduction is central to the proposed argument. Merely exhibiting the equal-half assignment would show only that normalization and complement consistency are compatible with nonadditivity. Theorem 1 asserts the stronger conclusion that every assignment satisfying the initial conditions must take this form. It is this universality claim that leads to the proposed exclusion of classical probability on richer event algebras.

The resulting failures share a common origin. Binomial aggregation adds weights over disjoint sample outcomes. Concentration and frequency limits require assignments to certain proper events to become arbitrarily small. A Gaussian limit requires threshold-event assignments to vary with the threshold. Under the derived rule, every proper event retains assignment $1/2$, so these conclusions fail by direct evaluation.

Classical independence is also incompatible with the derived assignment on repetitions. For $n \geq 2$, let $A = \{B_1 = 1\}$ and $B = \{B_2 = 1\}$. All

three events A , B , and $A \cap B$ are proper, giving

$$Q_n(A \cap B) = \frac{1}{2},$$

$$Q_n(A)Q_n(B) = \frac{1}{4}.$$

Thus the classical factorization condition cannot be imposed on these coordinates while retaining the universal assignment. The claimed challenge to the classical framework therefore reaches its independence structure as well as its limit formulas.

Local binary calculations survive because the restriction to the four-element algebra of a single question is additive. They do not supply a global expectation for arbitrary observables or justify summing assignments over a larger partition. This separation explains why the local moment and entropy identities coexist with the failure of global statistical aggregation.

10 Conclusion

Starting from normalization and complement consistency, the universal binary reduction identifies every proper-event assignment with one half. Its global consequences include failure of finite additivity, monotone continuity, classical independence, binomial aggregation, frequency concentration, and the standard Gaussian limit for standardized binary sums. Almost-sure frequency convergence likewise fails when evaluated on the infinite sequence space.

The proposed refutation of the classical framework rests on the universality of the reduction: the equal-half rule is presented as forced by binary structure, rather than offered merely as an alternative model. Every subsequent obstruction follows from that same rule. Binary normalization remains intact, while the global additive and asymptotic structure of classical probability is rejected.

Data availability

Demographic values are attributed in Table 1. The NASDAQ illustrations are supplied as the

three image files accompanying this source. The mathematical results require no numerical code.

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